

Data augmentation and contrastive learning

Matthew Engelhard

Duke Biostatistics & Bioinformatics

Last Time

- CNN basics
- CNN architectures

Today

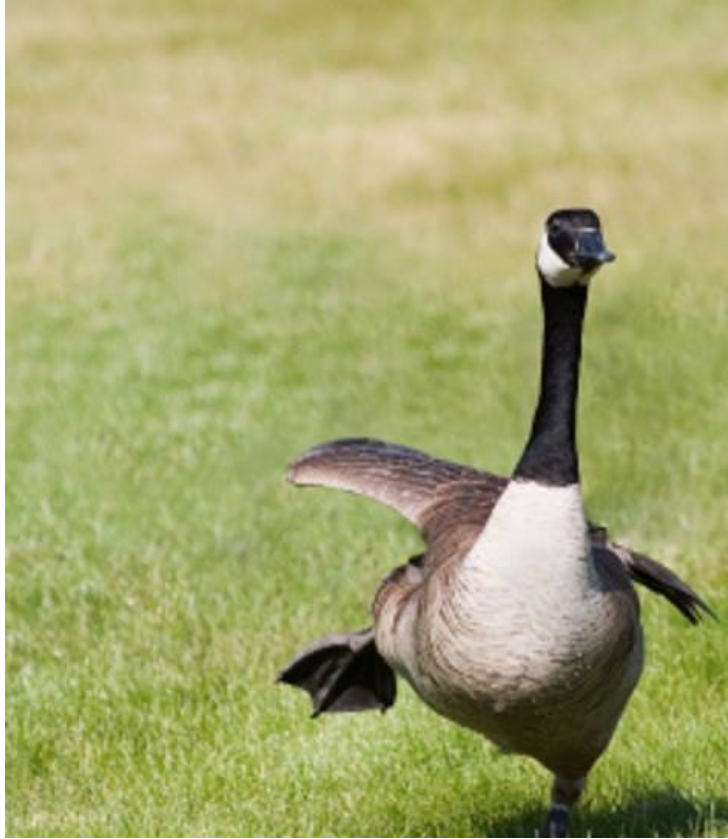
- Discussion of data augmentation for computer vision
- From data augmentation to contrastive learning
- *A Simple Framework for Contrastive Learning of Visual Representations* (SimCLR)
- Examples of contrastive learning in healthcare
- A look ahead at multi-modal data

Soon

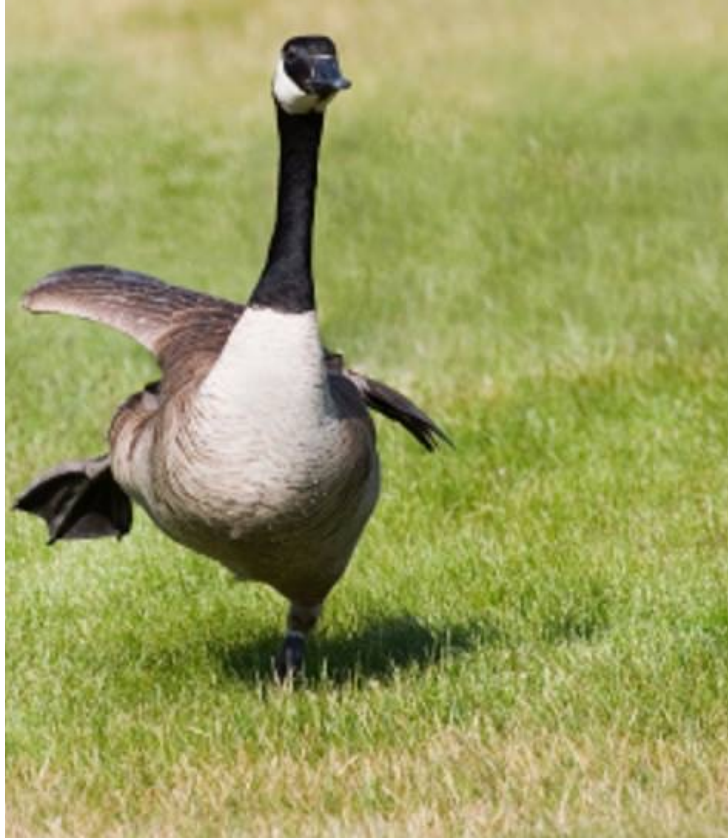
- Attribution methods / XAI for computer vision and beyond
- (activity) a closer look at contrastive learning
- Midterm on March 4

Data Augmentation

CNN: It's a goose



CNN: (translation invariance) Still a goose!



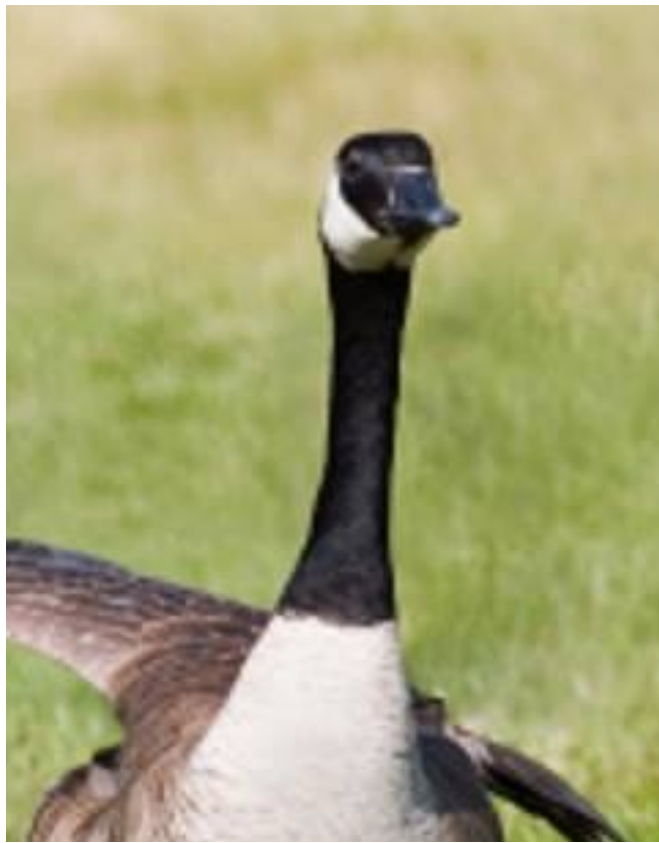
CNN: ?



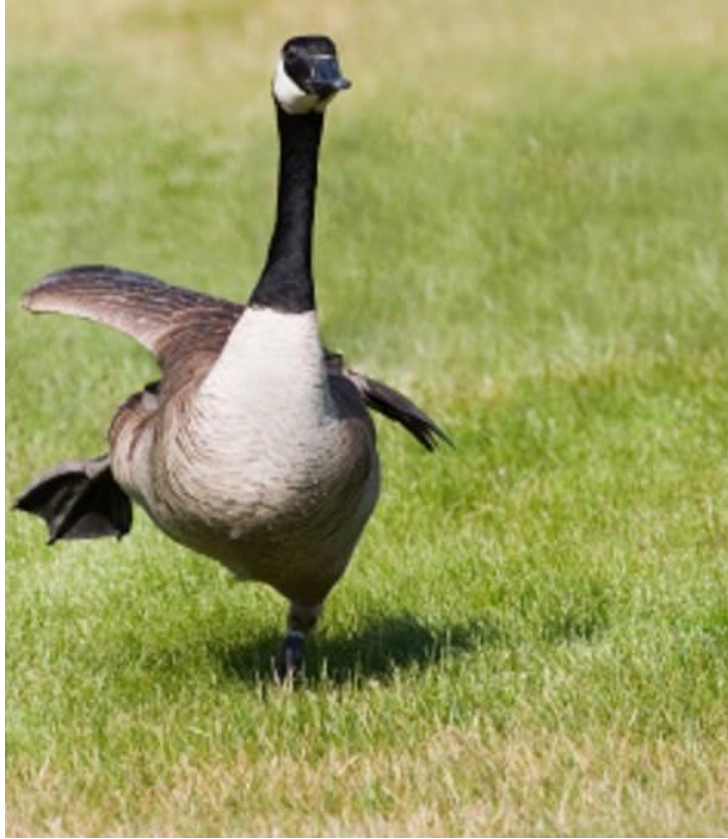
CNN: I don't see anything.



CNN: This is a bowling pin.

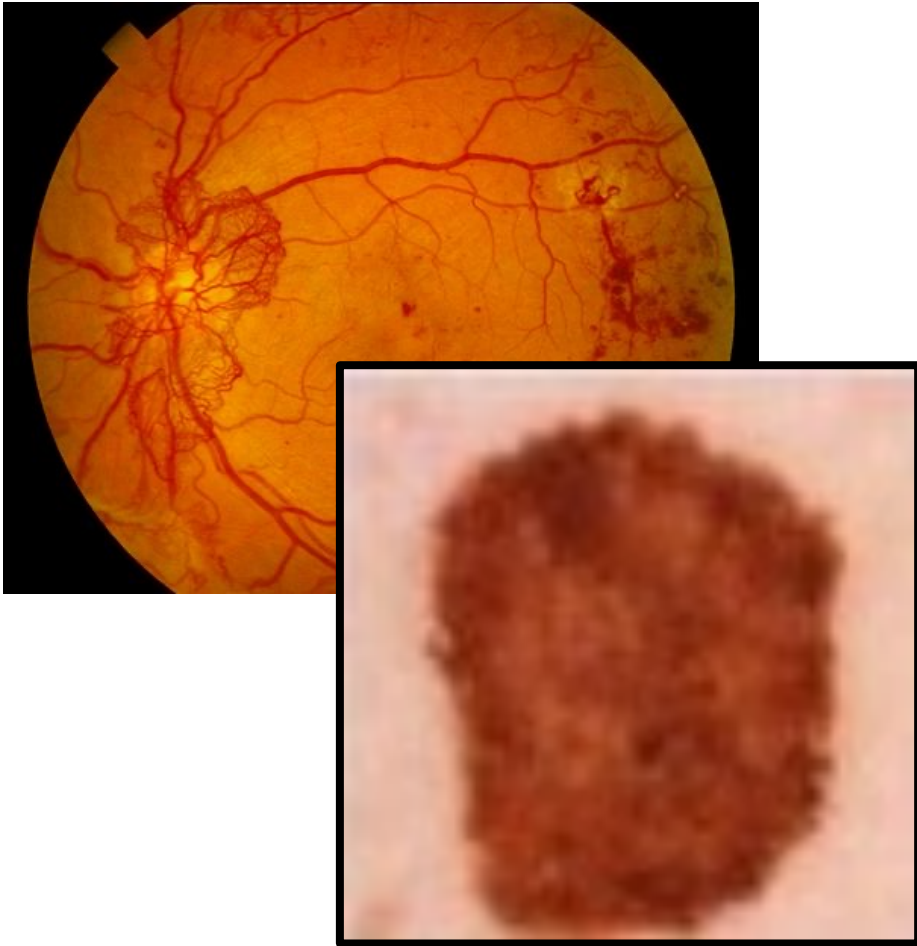


Data augmentation from first principles



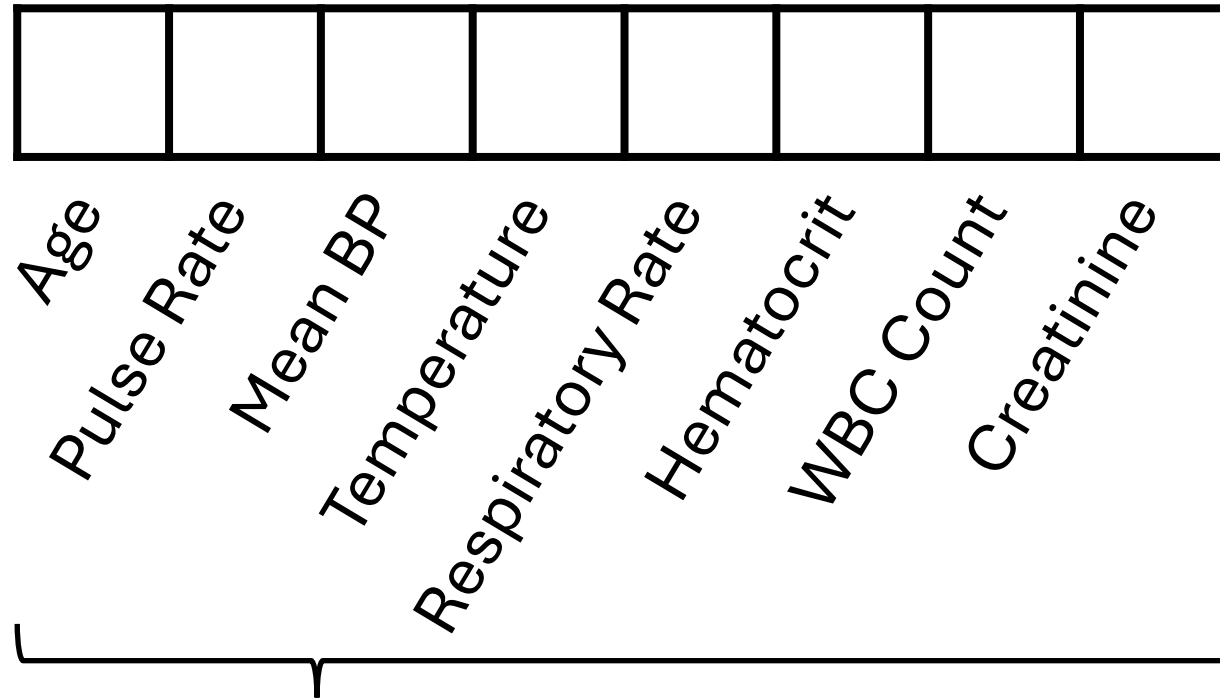
- What kind of changes do we want to be robust to?
- What data augmentation follows from this?

Data augmentation from first principles

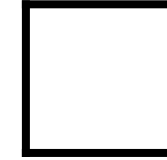


- What kind of changes do we want to be robust to?
- What data augmentation follows from this?
- What are key differences between medical images and naturalistic images?
- What are the implications for data augmentation?

Consider augmentations for tabular data



x , data/features for
a subject or patient



y , associated
value or label

Survival

**Are there analogous
augmentations?**

Why or why not?

Augmentation summary

- Consider what variations you expect your model to encounter in the real world / when generalizing
- Effective augmentation has a big impact on generalization performance

Contrastive Learning

From augmentation to contrastive learning

Consider the following line of reasoning:

1. Predictions should be robust to superficial augmentations, i.e. the prediction should not change
2. This should be true for all prediction tasks we might consider
3. This in turn implies that latent representations should be the same
4. Contrastive learning is motivated by this observation

This paradigm proves to be surprisingly effective.

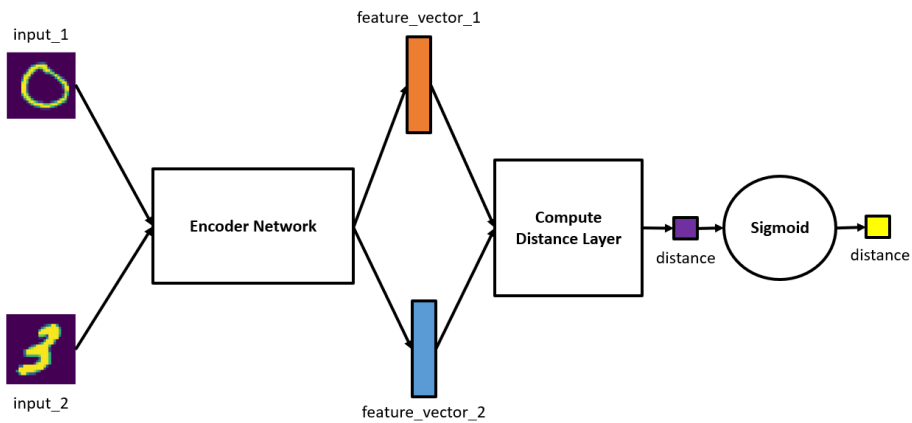
An early contrastive loss

- Simply penalizing the distance between views (i.e., augmentations) leads to mode collapse
- Positive pairs must therefore be contrasted against negative pairs
- An early version of this (circa 2005) was the “maximum margin contrastive loss”:

$$y d_{\theta}(\mathbf{x}_1, \mathbf{x}_2)^2 + (1 - y) [\max(0, m - d_{\theta}(\mathbf{x}_1, \mathbf{x}_2))]^2$$

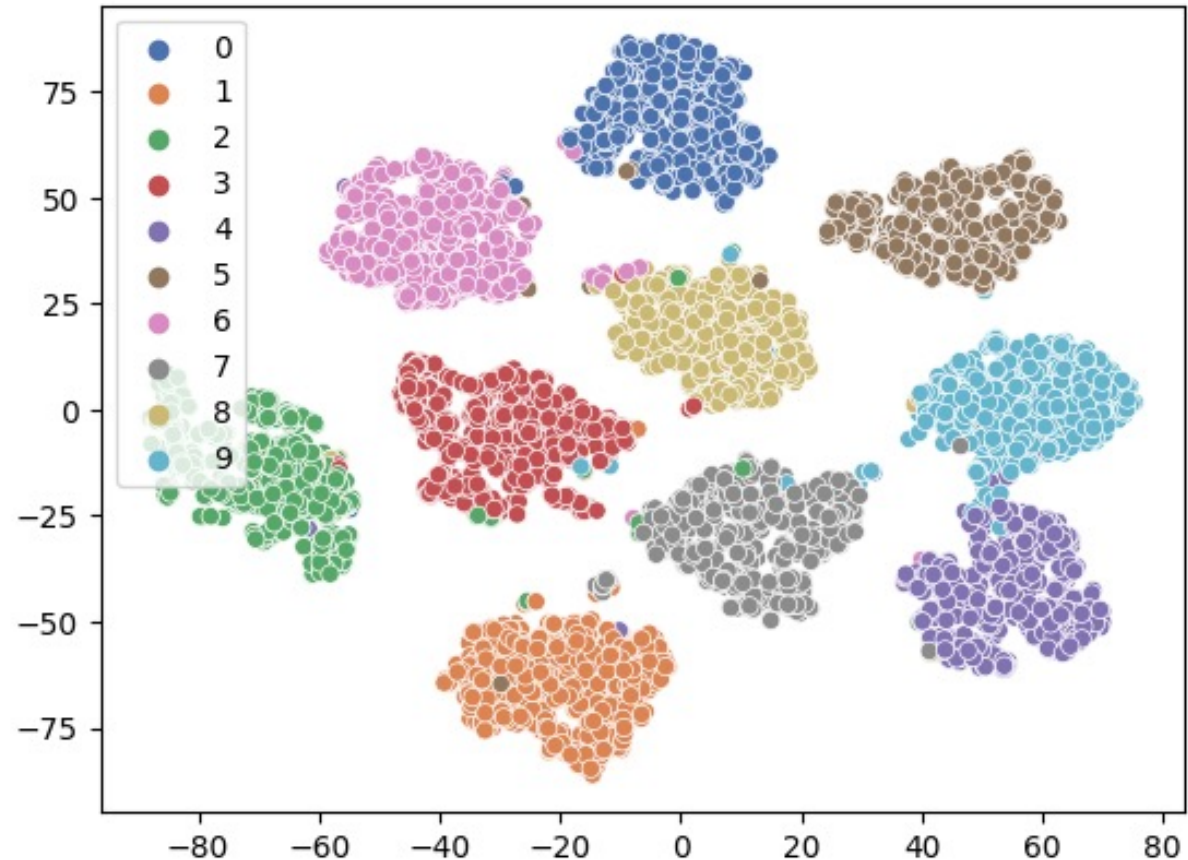
where d_{θ} is the distance in representation space after applying the encoder g_{θ} .

MNIST: digits form well-defined clusters



Maximum margin contrastive loss

$$y d_{\theta}(x_1, x_2)^2 + (1 - y)[\max(0, m - d_{\theta}(x_1, x_2))]^2$$



SimCLR: 3 primary improvements

- Choice of loss function
- Minibatch construction
- Use of a projection head

Minibatch construction and loss

- N examples each augmented in 2 ways -> minibatch size 2N
- Let $\text{sim}(\mathbf{u}, \mathbf{v})$ denote the cosine similarity $\mathbf{u}^T \mathbf{v} / \|\mathbf{u}\| \|\mathbf{v}\|$
- The loss function for a positive pair of examples (i, j) is given by:

$$-\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)}$$
$$= \underbrace{-\text{sim}(z_i, z_j)/\tau}_{\text{attraction between positive pairs}} + \underbrace{\log \left(\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau) \right)}_{\text{repulsion between negative pairs}}$$

“SimCLR” or “NT-Xent” (normalized temperature-scaled cross-entropy) loss.

Benefit of larger batch size

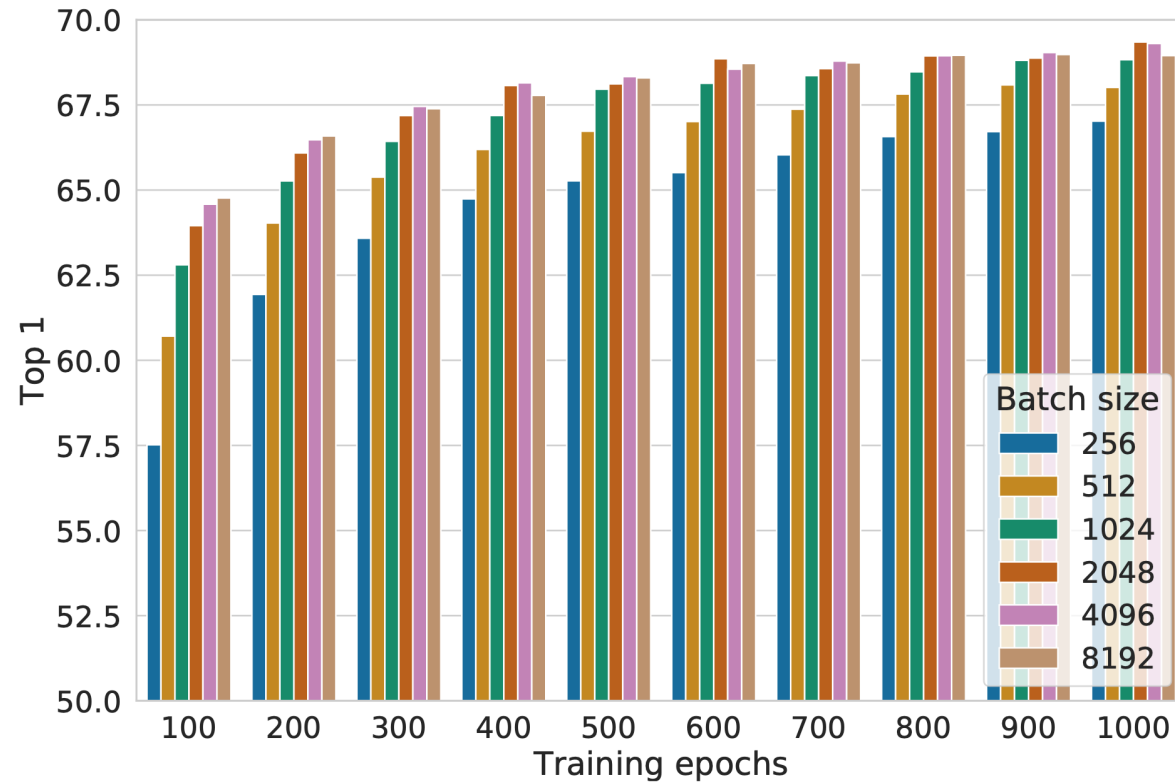


Figure 9. Linear evaluation models (ResNet-50) trained with different batch size and epochs. Each bar is a single run from scratch.¹⁰

Contrastive loss applied after projection head

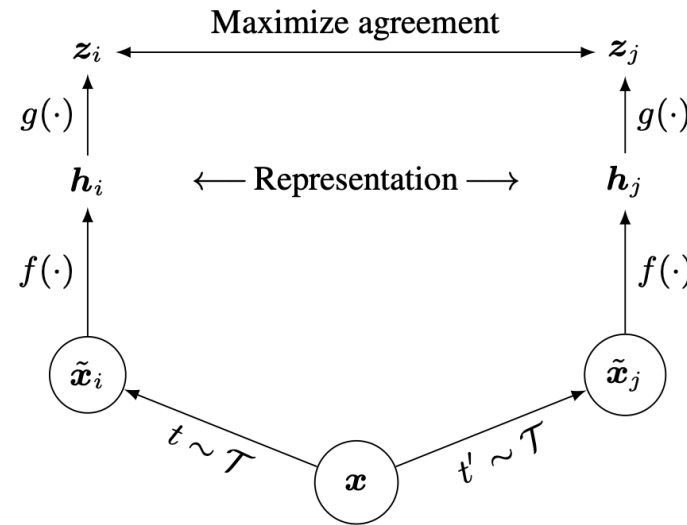
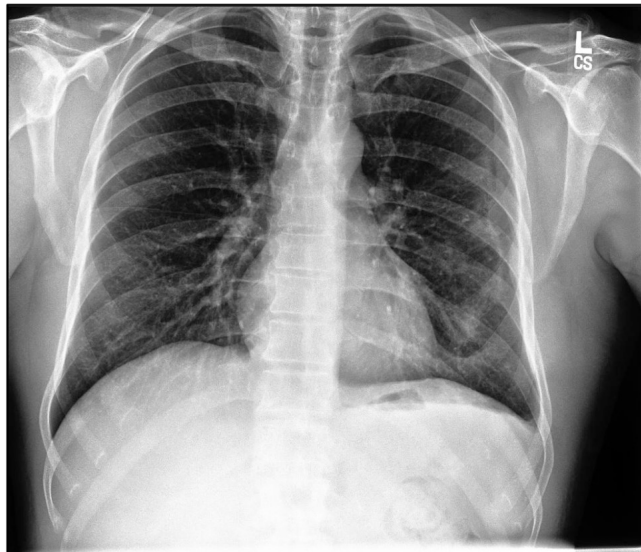


Figure 2. A simple framework for contrastive learning of visual representations. Two separate data augmentation operators are sampled from the same family of augmentations ($t \sim \mathcal{T}$ and $t' \sim \mathcal{T}$) and applied to each data example to obtain two correlated views. A base encoder network $f(\cdot)$ and a projection head $g(\cdot)$ are trained to maximize agreement using a contrastive loss. After training is completed, we throw away the projection head $g(\cdot)$ and use encoder $f(\cdot)$ and representation h for downstream tasks.

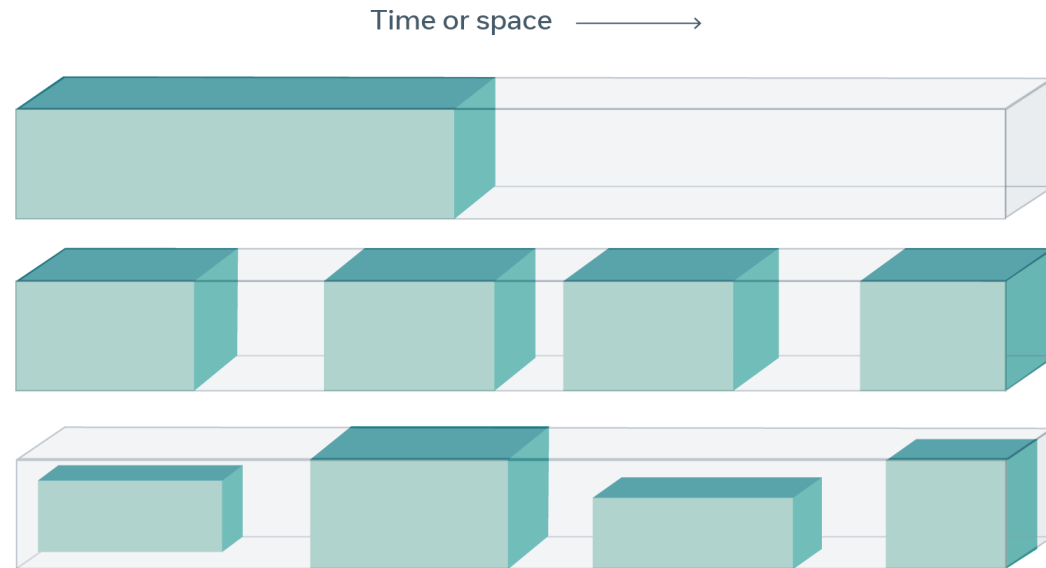
Contrastive learning in coding assignment 2

- Evaluate benefits of SimCLR-based pre-training compared to:
 - Full or partial fine-tuning
 - Using representations learned from ImageNet

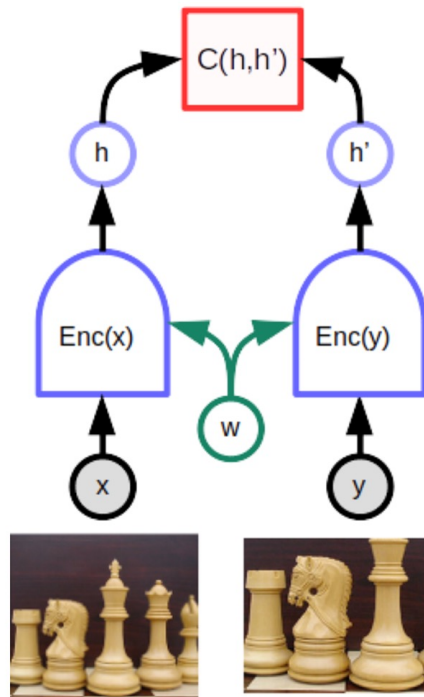


Self-supervised learning: The dark matter of intelligence

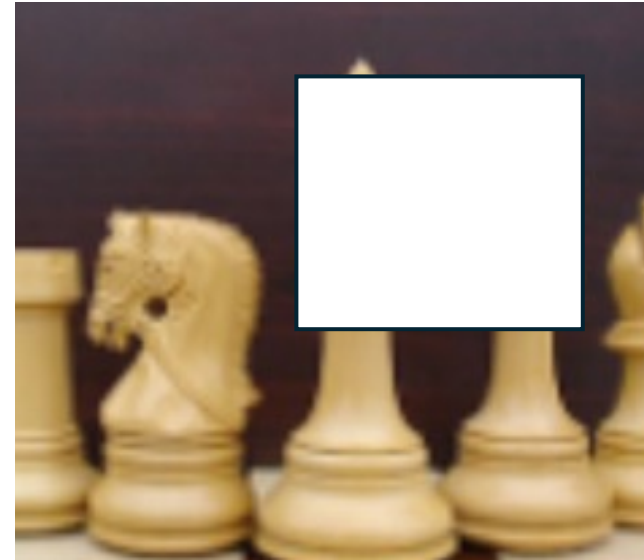
“Self-supervised learning obtains supervisory signals from the data itself, often leveraging the underlying structure in the data. The general technique of self-supervised learning is to predict any unobserved or hidden part (or property) of the input from any observed or unhidden part of the input.”



Two key variations on SSL



contrastive / noise contrastive



masked token / patch / sequence prediction

Examples

Contrastive clustering

- The instance-level contrastive head follows SimCLR
- The cluster-level contrastive head ensures cluster assignment is robust to augmentation
- This is surprisingly effective in practice, e.g. compared to earlier deep clustering methods

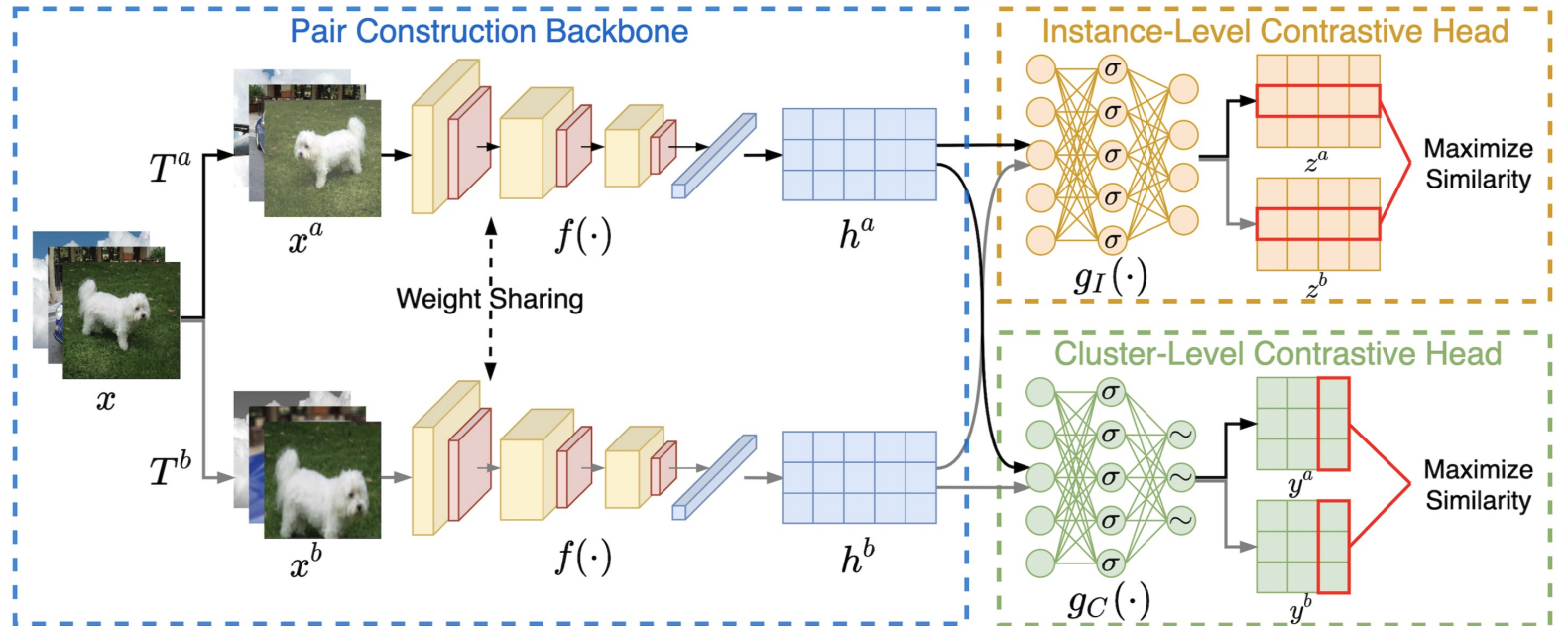
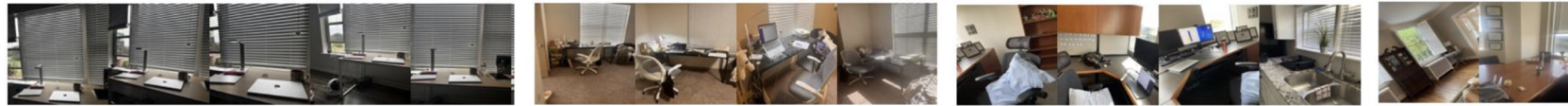


Fig. 2. The framework of Contrastive Clustering. We construct data pairs using two data augmentations. Given data pairs, one shared deep neural network is used to extract features from different augmentations. Two separate MLPs (σ denotes the *ReLU* activation and $\sim$ denotes the *Softmax* operation to produce soft labels) are used to project the features into the row and column space wherein the instance- and cluster-level contrastive learning are conducted respectively.

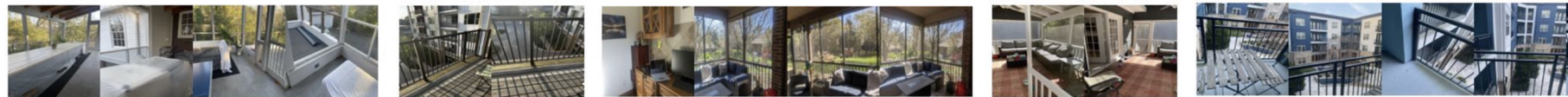
Infinite hierarchical contrastive clustering for personal digital envirotyping

— — — Same location type, *different* participants — — —

Work area



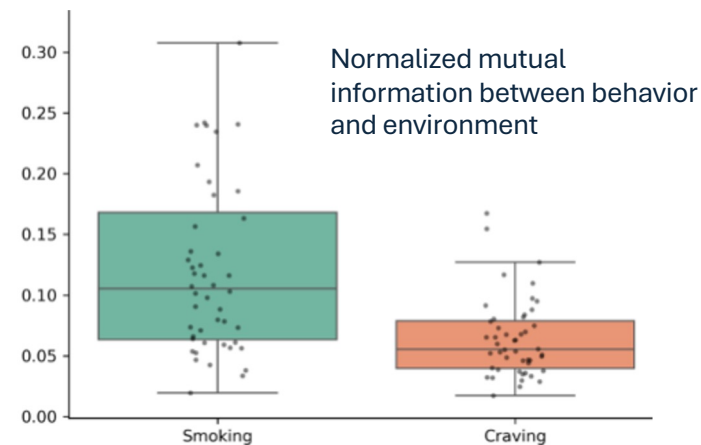
Porch



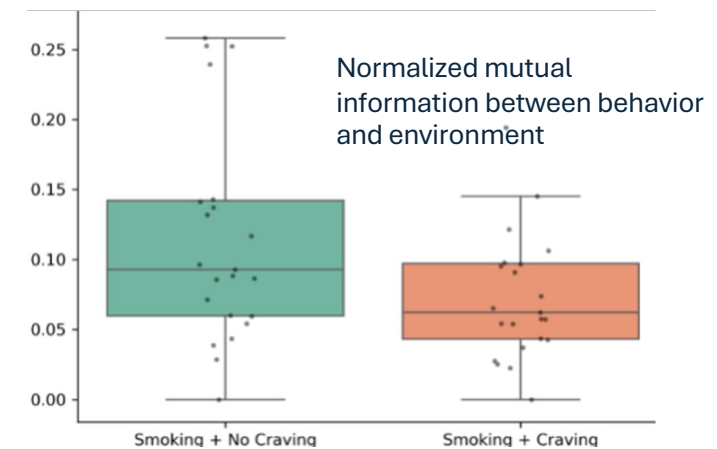
Eating area



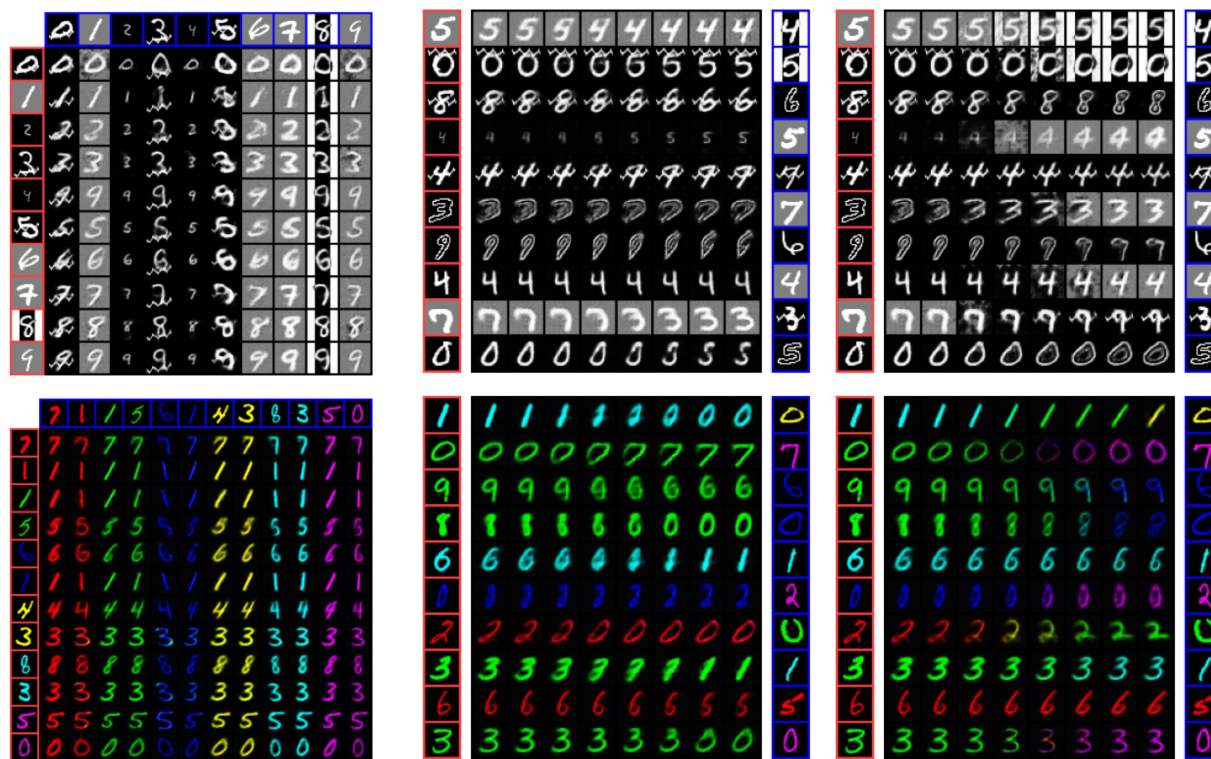
The act of smoking is more environmentally linked than the urge to smoke.



The importance of the environment is reduced when urge is high.



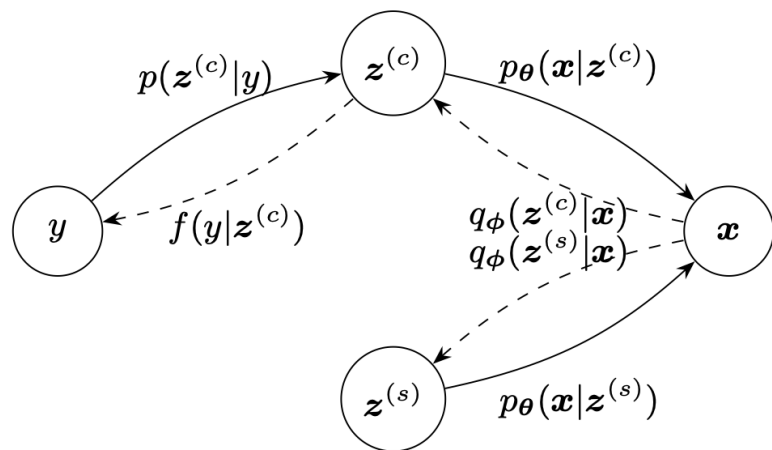
Disentangling style and content



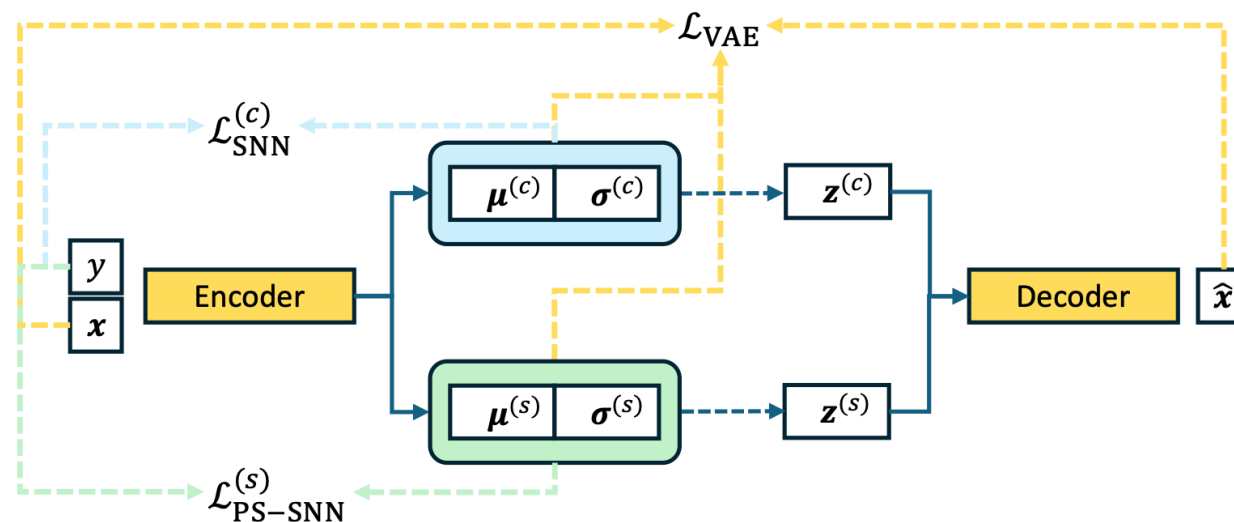
Sun, Minghui, Benjamin A. Goldstein, and Matthew M. Engelhard. "CLEAR: Unlearning Spurious Style-Content Associations with Contrastive LEarning with Anti-contrastive Regularization." *arXiv preprint arXiv:2507.18794* (2025).

Figure 4: Swapping and interpolation experiments. In (a), each image is generated using the content semantics from the red axis and style semantics from the blue axis. (b) interpolates content and (c) interpolates style from the source to the target sample, keeping the other factor fixed.

Disentangling style and content



(a) graphical representation



(b) schematic illustration

Figure 2: Graphical representation and schematic illustration of CLEAR-VAE

- Standard contrastive loss (NT-Xent) maximizes mutual information between label and content features
- Anti-contrastive loss minimizes the mutual information between label and style features

Contrastive learning for tabular data

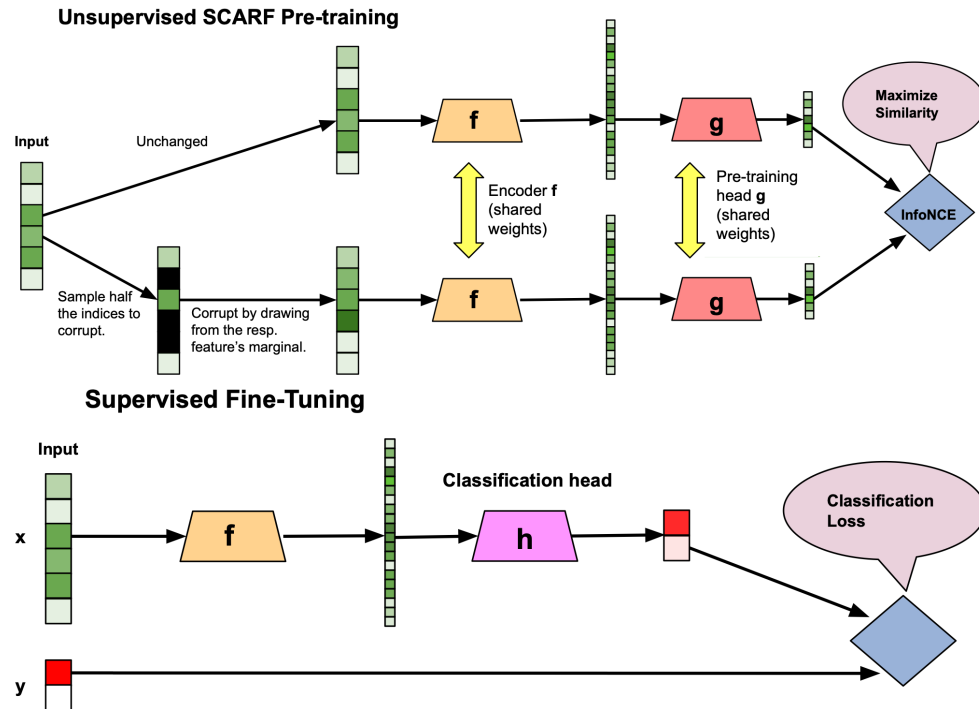
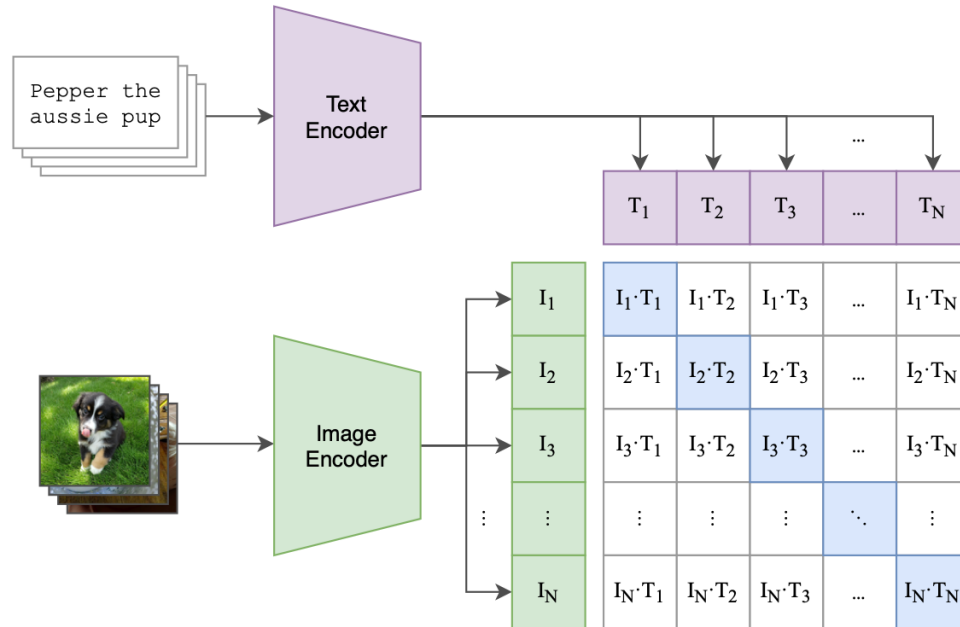


Figure 1: Diagram showing unsupervised SCARF pre-training (**Top**) and subsequent supervised fine-tuning (**Bottom**). During pre-training, networks f and g are learned to produce good representations of the input data. After pre-training, g is discarded and a classification head h is applied on top of the learned f and both f and h are subsequently fine-tuned for classification.

- Self-supervised representation learning for tabular data
- Does improve performance in our experiments
- Easy to implement

A look ahead at (joint) multi-modal fusion

(1) Contrastive pre-training



(2) Create dataset classifier from label text

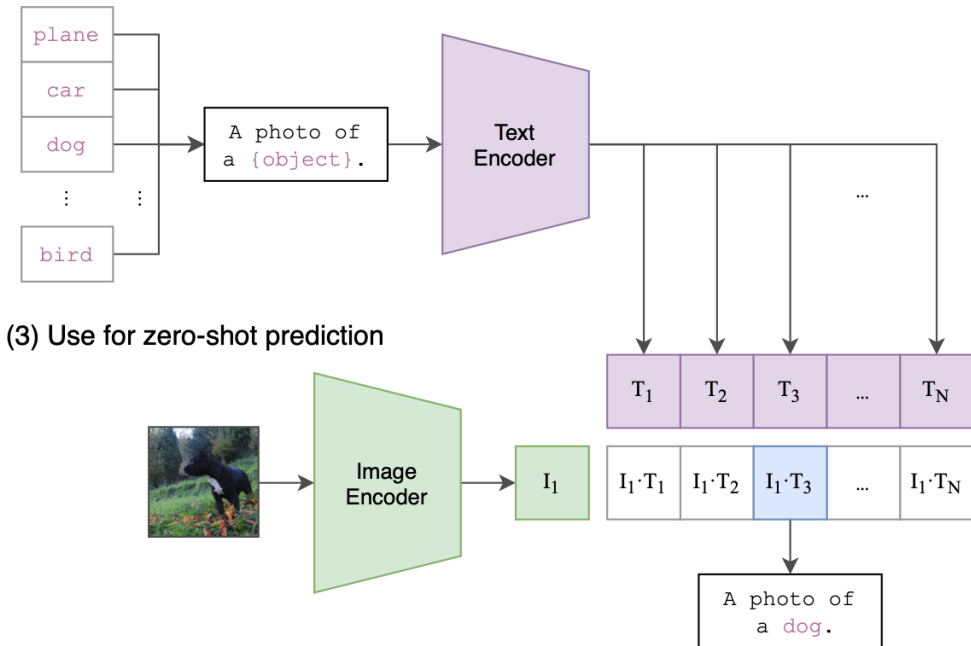


Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.